

## Assignment-2(1)

Load dataset into a SQL database.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'sakila', 'shopbase', and 'superstore\_analysis'. The 'superstore\_analysis' schema is selected, showing its tables, views, stored procedures, and functions. The main editor window contains the following SQL script:

```
1 CREATE DATABASE superstore_analysis;
2 USE superstore_analysis;
3
4 SHOW tables;
5
6 SELECT * FROM superstore
```

The 'Result Grid' shows the output of the 'SHOW tables' query, listing 10 tables in the 'superstore' database. The tables are: superstore, superstore2, superstore3, superstore4, superstore5, superstore6, superstore7, superstore8, superstore9, and superstore10. The 'superstore' table is highlighted.

| Row ID | Order ID       | Order Date | Ship Date  | Ship Mode      | Customer ID | Customer Name   | Segment   | Country       | City            | State      | Postal Code | Region | Product |
|--------|----------------|------------|------------|----------------|-------------|-----------------|-----------|---------------|-----------------|------------|-------------|--------|---------|
| 1      | CA-2016-152156 | 11/8/2016  | 11/11/2016 | Second Class   | CG-12520    | Claire Gite     | Consumer  | United States | Henderson       | Kentucky   | 42420       | South  | FUR-1   |
| 2      | CA-2016-152156 | 11/8/2016  | 11/11/2016 | Second Class   | CG-12520    | Claire Gite     | Consumer  | United States | Henderson       | Kentucky   | 42420       | South  | FUR-1   |
| 3      | CA-2016-138688 | 6/12/2016  | 6/16/2016  | Second Class   | DV-13045    | Darrin Van Huff | Corporate | United States | Los Angeles     | California | 90036       | West   | OFF-1   |
| 4      | US-2015-108966 | 10/11/2015 | 10/18/2015 | Standard Class | SO-20335    | Sean O'Donnell  | Consumer  | United States | Fort Lauderdale | Florida    | 33311       | South  | FUR-2   |
| 5      | US-2015-108966 | 10/11/2015 | 10/18/2015 | Standard Class | SO-20335    | Sean O'Donnell  | Consumer  | United States | Fort Lauderdale | Florida    | 33311       | South  | OFF-1   |
| 6      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer  | United States | Los Angeles     | California | 90032       | West   | FUR-1   |
| 7      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer  | United States | Los Angeles     | California | 90032       | West   | OFF-1   |
| 8      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer  | United States | Los Angeles     | California | 90032       | West   | TEC-1   |
| 9      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer  | United States | Los Angeles     | California | 90032       | West   | OFF-1   |
| 10     | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer  | United States | Los Angeles     | California | 90032       | West   | OFF-1   |

Apply WHERE filters (region, category, date, sales)

The screenshot shows the MySQL Workbench interface with the same 'superstore\_analysis' schema. The main editor window contains the following SQL script:

```
10
11 SELECT *
12 FROM superstore
13 WHERE region='West';
14
```

The 'Result Grid' shows the output of the query, listing 10 rows of data from the 'superstore' table. The rows are filtered by the 'WHERE region='West'' condition. The tables are: superstore, superstore2, superstore3, superstore4, superstore5, superstore6, superstore7, superstore8, superstore9, and superstore10. The 'superstore' table is highlighted.

| Row ID | Order ID       | Order Date | Ship Date  | Ship Mode      | Customer ID | Customer Name   | Segment   | Country       | City        | State      | Postal Code | Region | Product |
|--------|----------------|------------|------------|----------------|-------------|-----------------|-----------|---------------|-------------|------------|-------------|--------|---------|
| 3      | CA-2016-138688 | 6/12/2016  | 6/16/2016  | Second Class   | DV-13045    | Darrin Van Huff | Corporate | United States | Los Angeles | California | 90036       | West   | OFF-1   |
| 6      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer  | United States | Los Angeles | California | 90032       | West   | FUR-1   |
| 7      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer  | United States | Los Angeles | California | 90032       | West   | OFF-1   |
| 8      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer  | United States | Los Angeles | California | 90032       | West   | TEC-1   |
| 9      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer  | United States | Los Angeles | California | 90032       | West   | OFF-1   |
| 10     | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer  | United States | Los Angeles | California | 90032       | West   | OFF-1   |
| 11     | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer  | United States | Los Angeles | California | 90032       | West   | FUR-1   |
| 12     | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer  | United States | Los Angeles | California | 90032       | West   | TEC-1   |
| 14     | CA-2016-161389 | 12/5/2016  | 12/10/2016 | Standard Class | IM-15070    | Irene Maddox    | Consumer  | United States | Seattle     | Washington | 98103       | West   | OFF-1   |
| 18     | CA-2014-167164 | 5/13/2014  | 5/15/2014  | Second Class   | AG-10270    | Alejandro Grove | Consumer  | United States | West Jordan | Utah       | 84084       | West   | OFF-1   |

MySQL Workbench

New Connection x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

sakila  
 shopbase  
 **superstore\_analysis**  
 Tables  
 superstore  
 Views  
 Stored Procedures  
 Functions  
 sys  
 world

Administration Schemas

Information

Schema: **superstore\_analysis**

SQL File 3 x

Limit to 1000 rows

```
14  
15 SELECT *  
16 FROM superstore  
17 WHERE category='Technology';  
18
```

Result Grid

| Row ID | Order ID       | Order Date | Ship Date  | Ship Mode      | Customer ID | Customer Name      | Segment     | Country       | City          | State      | Postal Code | Region  | P  |
|--------|----------------|------------|------------|----------------|-------------|--------------------|-------------|---------------|---------------|------------|-------------|---------|----|
| 8      | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman    | Consumer    | United States | Los Angeles   | California | 90032       | West    | TE |
| 12     | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman    | Consumer    | United States | Los Angeles   | California | 90032       | West    | TE |
| 20     | CA-2014-143336 | 8/27/2014  | 9/1/2014   | Second Class   | ZD-21925    | Zuschuss Donatelli | Consumer    | United States | San Francisco | California | 94109       | West    | TE |
| 27     | CA-2016-121755 | 1/16/2016  | 1/20/2016  | Second Class   | BH-13945    | Eric Hoffmann      | Consumer    | United States | Los Angeles   | California | 90049       | West    | TE |
| 36     | CA-2016-117990 | 12/8/2016  | 12/10/2016 | First Class    | GH-14485    | Gene Hale          | Corporate   | United States | Richardson    | Texas      | 75080       | Central | TE |
| 41     | CA-2015-117415 | 12/27/2015 | 12/31/2015 | Standard Class | SN-20710    | Steve Nguyen       | Home Office | United States | Houston       | Texas      | 77041       | Central | TE |
| 42     | CA-2017-120999 | 9/10/2017  | 9/15/2017  | Standard Class | LC-16930    | Linda Cazamas      | Corporate   | United States | Naperville    | Illinois   | 60540       | Central | TE |
| 45     | CA-2016-118255 | 3/11/2016  | 3/13/2016  | First Class    | ON-18715    | Odella Nelson      | Corporate   | United States | Eagan         | Minnesota  | 55122       | Central | TE |
| 48     | CA-2016-169194 | 6/20/2016  | 6/25/2016  | Standard Class | LH-16900    | Lena Hernandez     | Consumer    | United States | Dover         | Delaware   | 19901       | East    | TE |
| 49     | CA-2016-169194 | 6/20/2016  | 6/25/2016  | Standard Class | LH-16900    | Lena Hernandez     | Consumer    | United States | Dover         | Delaware   | 19901       | East    | TE |

superstore 5 x

Read Only

Output

Action Output

| #  | Time     | Action                                                             | Message                                                           | Duration / Fetch      |
|----|----------|--------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------|
| 21 | 23.45:59 | SELECT * FROM sales WHERE category='Technology' LIMIT 0, 1000      | Error Code: 1146. Table 'superstore_analysis sales' doesn't exist | 0.000 sec             |
| 22 | 21.46:14 | SELECT * FROM superstore WHERE category='Technology' LIMIT 0, 1000 | 1000 row(s) returned                                              | 0.000 sec / 0.015 sec |

MySQL Workbench

New Connection x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

sakila  
 shopbase  
 **superstore\_analysis**  
 Tables  
 superstore  
 Views  
 Stored Procedures  
 Functions  
 sys  
 world

Administration Schemas

Information

Schema: **superstore\_analysis**

SQL File 3 x

Limit to 1000 rows

```
17 WHERE category='Technology';  
18  
19 SELECT *  
20 FROM superstore  
21 WHERE sales > 10000;
```

Result Grid

| Row ID | Order ID       | Order Date | Ship Date  | Ship Mode      | Customer ID | Customer Name   | Segment   | Country       | City          | State        | Postal Code | Region  | Pr  |
|--------|----------------|------------|------------|----------------|-------------|-----------------|-----------|---------------|---------------|--------------|-------------|---------|-----|
| 11     | CA-2014-115812 | 6/9/2014   | 6/14/2014  | Standard Class | BH-11710    | Brosina Hoffman | Consumer  | United States | Los Angeles   | California   | 90032       | West    | FUI |
| 25     | CA-2015-106320 | 9/25/2015  | 9/30/2015  | Standard Class | EB-13870    | Emily Burns     | Consumer  | United States | Orem          | Utah         | 84057       | West    | FUI |
| 28     | US-2015-150630 | 9/17/2015  | 9/21/2015  | Standard Class | TB-21520    | Tracy Blumstein | Consumer  | United States | Philadelphia  | Pennsylvania | 19140       | East    | FUI |
| 36     | CA-2016-117990 | 12/8/2016  | 12/10/2016 | First Class    | GH-14485    | Gene Hale       | Corporate | United States | Richardson    | Texas        | 75080       | Central | TE  |
| 55     | CA-2016-105816 | 12/11/2016 | 12/17/2016 | Standard Class | JM-15265    | Janet Molinari  | Corporate | United States | New York City | New York     | 10024       | East    | TE  |
| 68     | CA-2014-106376 | 12/5/2014  | 12/10/2014 | Standard Class | BS-11590    | Brendan Sweed   | Corporate | United States | Gilbert       | Arizona      | 85234       | West    | OF  |
| 150    | CA-2016-114489 | 12/5/2016  | 12/9/2016  | Standard Class | JE-16165    | Justin Ellison  | Corporate | United States | Franklin      | Wisconsin    | 53132       | Central | FUI |
| 166    | CA-2014-139892 | 9/8/2014   | 9/12/2014  | Standard Class | BM-11140    | Becky Martin    | Consumer  | United States | San Antonio   | Texas        | 78207       | Central | TE  |
| 168    | CA-2014-139892 | 9/8/2014   | 9/12/2014  | Standard Class | BM-11140    | Becky Martin    | Consumer  | United States | San Antonio   | Texas        | 78207       | Central | FUI |
| 216    | CA-2015-146262 | 1/2/2015   | 1/9/2015   | Standard Class | VW-21775    | Victoria Wilson | Corporate | United States | Medina        | Ohio         | 44256       | East    | TE  |

superstore 6 x

Read Only

Output

Action Output

| #  | Time     | Action                                                             | Message              | Duration / Fetch      |
|----|----------|--------------------------------------------------------------------|----------------------|-----------------------|
| 22 | 23.46:14 | SELECT * FROM superstore WHERE category='Technology' LIMIT 0, 1000 | 1000 row(s) returned | 0.000 sec / 0.016 sec |
| 23 | 21.46:46 | SELECT * FROM superstore WHERE sales > 10000 LIMIT 0, 1000         | 467 row(s) returned  | 0.000 sec / 0.015 sec |

## Use GROUP BY for aggregations (sales, quantity, averages).

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'superstore\_analysis' selected. The main editor contains the following SQL query:

```
22
23 • SELECT Region,
24       SUM(Sales) AS Total_Sales
25 FROM Superstore
26 GROUP BY Region;
```

The 'Result Grid' shows the output of the query:

| Region  | Total_Sales        |
|---------|--------------------|
| South   | 388983.58500000037 |
| West    | 713471.34450000004 |
| Central | 497800.8728000007  |
| East    | 672194.05399999981 |

The 'Output' pane shows the execution log:

| #  | Time     | Action                                                                     | Message                                                       | Duration / Fetch      |
|----|----------|----------------------------------------------------------------------------|---------------------------------------------------------------|-----------------------|
| 24 | 23:47:28 | SELECT Region, SUM(superstore) AS total_sales FROM superstore GROUP BY ... | Error Code: 1054. Unknown column 'superstore' in 'field list' | 0.000 sec             |
| 25 | 23:49:03 | SELECT Region, SUM(Sales) AS Total_Sales FROM Superstore GROUP BY ...      | 4 row(s) returned                                             | 0.015 sec / 0.000 sec |

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'superstore\_analysis' selected. The main editor contains the following SQL query:

```
27
28 • SELECT Category,
29       SUM(Quantity) AS Total_Quantity
30 FROM Superstore
31 GROUP BY Category;
```

The 'Result Grid' shows the output of the query:

| Category        | Total_Quantity |
|-----------------|----------------|
| Furniture       | 7855           |
| Office Supplies | 21990          |
| Technology      | 6904           |

The 'Output' pane shows the execution log:

| #  | Time     | Action                                                                   | Message           | Duration / Fetch      |
|----|----------|--------------------------------------------------------------------------|-------------------|-----------------------|
| 25 | 23:49:03 | SELECT Region, SUM(Sales) AS Total_Sales FROM Superstore GROUP BY ...    | 4 row(s) returned | 0.015 sec / 0.000 sec |
| 26 | 23:49:27 | SELECT Category, SUM(Quantity) AS Total_Quantity FROM Superstore GROU... | 3 row(s) returned | 0.016 sec / 0.000 sec |

MySQL Workbench

New Connection x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

sakila  
shopease  
superstore\_analysis

Tables  
superstore  
Views  
Stored Procedures  
Functions

sys  
world

Administration Schemas

Information

Schema: superstore\_analysis

SQL File 3\* x

Limit to 1000 rows

```
32  
33 • SELECT Category,  
34     AVG(Sales) AS Average_Sales  
35 FROM Superstore  
36 GROUP BY Category;
```

Result Grid

| Category        | Average_Sales      |
|-----------------|--------------------|
| Furniture       | 353.4459311957568  |
| Office Supplies | 121.69225531914947 |
| Technology      | 454.5405475802047  |

Result 9 x

Read Only

Output

Action Output

| #  | Time     | Action                                                                              | Message           | Duration / Fetch      |
|----|----------|-------------------------------------------------------------------------------------|-------------------|-----------------------|
| 26 | 23:49:27 | SELECT Category, SUM(Quantity) AS Total_Quantity FROM Superstore GROUP BY Category; | 3 row(s) returned | 0.016 sec / 0.000 sec |
| 27 | 23:49:46 | SELECT Category, AVG(Sales) AS Average_Sales FROM Superstore GROUP BY Category;     | 3 row(s) returned | 0.016 sec / 0.000 sec |

Object Info Session

MySQL Workbench

New Connection x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

sakila  
shopease  
superstore\_analysis

Tables  
superstore  
Views  
Stored Procedures  
Functions

sys  
world

Administration Schemas

Information

Schema: superstore\_analysis

SQL File 3\* x

Limit to 1000 rows

```
37  
38 • SELECT Category,  
39     SUM(Profit) AS Total_Profit  
40 FROM Superstore  
41 GROUP BY Category;
```

Result Grid

| Category        | Total_Profit       |
|-----------------|--------------------|
| Furniture       | 16980.771199999998 |
| Office Supplies | 120489.88640000001 |
| Technology      | 145387.09659999999 |

Result 10 x

Read Only

Output

Action Output

| #  | Time     | Action                                                                          | Message           | Duration / Fetch      |
|----|----------|---------------------------------------------------------------------------------|-------------------|-----------------------|
| 27 | 23:49:46 | SELECT Category, AVG(Sales) AS Average_Sales FROM Superstore GROUP BY Category; | 3 row(s) returned | 0.016 sec / 0.000 sec |
| 28 | 23:50:15 | SELECT Category, SUM(Profit) AS Total_Profit FROM Superstore GROUP BY Category; | 3 row(s) returned | 0.016 sec / 0.000 sec |

Object Info Session

MySQL Workbench

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

sakila  
 shopease  
 superstore\_analysis  
   Tables  
     superstore  
   Views  
   Stored Procedures  
   Functions  
 sys  
 world

Administration Schemas

Information

Schema: superstore\_analysis

SQL File 3

```

42
43 SELECT Region,
44        COUNT(*) AS Orders_Count,
45        SUM(Sales) AS Revenue,
46        SUM(Profit) AS Profit
47 FROM Superstore
48 GROUP BY Region
49 ORDER BY Revenue DESC;

```

Result Grid

| Region  | Orders_Count | Revenue            | Profit             |
|---------|--------------|--------------------|--------------------|
| West    | 3099         | 713471.3445000004  | 106021.14950000007 |
| East    | 2756         | 672194.0539999981  | 90672.01270000022  |
| Central | 2260         | 497800.8728000007  | 40128.903          |
| South   | 1579         | 388983.58500000037 | 46035.68900000004  |

Result 11

Output

Action Output

| #  | Time     | Action                                                                | Message                                                       | Duration / Fetch      |
|----|----------|-----------------------------------------------------------------------|---------------------------------------------------------------|-----------------------|
| 30 | 23:50:54 | SELECT Product_Name, SUM(Sales) AS Total_Sales FROM Superstore GRO... | Error Code: 1054. Unknown column 'Product_Name' in field list | 0.000 sec             |
| 31 | 23:52:51 | SELECT Region, COUNT(*) AS Orders_Count, SUM(Sales) AS Revenue, ...   | 4 row(s) returned                                             | 0.016 sec / 0.000 sec |

Object Info Session

Sort and limit results (top products, top categories).

MySQL Workbench

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

City  
 State  
 Postal Code  
 Region  
 Product ID  
 Category  
 Sub-Category  
 Product Name  
 Sales  
 Quantity  
 Discount  
 Profit  
 Indexes

Administration Schemas

Information

Table: superstore

Columns:

| Column     | Type |
|------------|------|
| Row ID     | int  |
| Order ID   | text |
| Order Date | text |
| Ship Date  | text |
| Ship Mode  | text |
| Customer   | text |
| ID         | text |
| Customer   | text |
| Name       | text |

SQL File 3

```

51 SELECT 'Product Name',
52        SUM(Sales) AS Total_Sales
53 FROM Superstore
54 GROUP BY 'Product Name'
55 ORDER BY Total_Sales DESC
56 LIMIT 10;

```

Result Grid

| Product Name                                       | Total_Sales        |
|----------------------------------------------------|--------------------|
| Canon imageCLASS 2200 Advanced Copier              | 61599.824          |
| Fellowes PB500 Electric Punch Plastic Comb Bind... | 27453.384          |
| Cisco TelePresence System EX90 Videoconferen...    | 22638.48           |
| HON 5400 Series Task Chairs for Big and Tall       | 21870.576          |
| GBC DocuBind TL300 Electric Binding System         | 19823.479000000003 |
| GBC Ibmaster 500 Manual ProClick Binding System    | 19024.5            |
| Hewlett Packard LaserJet 3310 Copier               | 18839.686          |
| HP DesignJet T520 Inkjet Large Format Printer ...  | 18374.895          |
| GBC DocuBind P400 Electric Binding System          | 17965.068          |
| Link Speed Automatic Electric Letter Opener        | 17030.310000000004 |

Result 12

Output

Action Output

| #  | Time     | Action                                                                  | Message                                                   | Duration / Fetch      |
|----|----------|-------------------------------------------------------------------------|-----------------------------------------------------------|-----------------------|
| 32 | 23:53:22 | SELECT Order_ID, COUNT(*) AS Duplicate_Count FROM Superstore GROU...    | Error Code: 1054. Unknown column 'Order_ID' in field list | 0.000 sec             |
| 33 | 00:04:42 | SELECT 'Product Name', SUM(Sales) AS Total_Sales FROM Superstore GRO... | 10 row(s) returned                                        | 0.031 sec / 0.000 sec |

Object Info Session



MySQL Workbench

New Connection x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- City
- State
- Postal Code
- Region
- Product ID
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit
- Indexes

Administration Schemas

Information

Table: **superstore**

Columns:

- Row ID int
- Order ID text
- Order Date text
- Ship Date text
- Ship Mode text
- Customer text
- ID text
- Customer Name text

Object Info Session

SQL File 3 x

Limit to 1000 rows

```

56 LIMIT 10;
57
58 SELECT 'Customer Name',
59 SUM(Sales) AS Total_Sales
60 FROM Superstore
61 GROUP BY 'Customer Name'
62 ORDER BY Total_Sales DESC

```

Result Grid

| Customer Name | Total_Sales        |
|---------------|--------------------|
| Sean Miller   | 25043.05           |
| Tamara Chand  | 19017.847999999998 |
| Raymond Buch  | 15117.339          |
| Tom Ashbrook  | 14595.62           |
| Adrian Barton | 14355.610999999997 |
| Sanjit Chand  | 14142.333999999999 |
| Ken Lonsdale  | 14071.917          |
| Hunter Lopez  | 12873.297999999999 |
| Sanjit Engle  | 12209.438000000002 |

Result 13 x

Output

Action Output

| #  | Time     | Action                                                                  | Message            | Duration / Fetch      |
|----|----------|-------------------------------------------------------------------------|--------------------|-----------------------|
| 33 | 00:04:42 | SELECT 'Product Name', SUM(Sales) AS Total_Sales FROM Superstore GRO... | 10 row(s) returned | 0.031 sec / 0.000 sec |
| 34 | 00:06:03 | SELECT 'Customer Name', SUM(Sales) AS Total_Sales FROM Superstore GR... | 10 row(s) returned | 0.031 sec / 0.000 sec |

MySQL Workbench

New Connection x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- City
- State
- Postal Code
- Region
- Product ID
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit
- Indexes

Administration Schemas

Information

Table: **superstore**

Columns:

- Row ID int
- Order ID text
- Order Date text
- Ship Date text
- Ship Mode text
- Customer text
- ID text
- Customer Name text

Object Info Session

SQL File 3 x

Limit to 1000 rows

```

65 SELECT 'Order ID',
66 COUNT(*) AS Duplicate_Count
67 FROM Superstore
68 GROUP BY 'Order ID'
69 HAVING COUNT(*) > 1;
70

```

Result Grid

| Order ID       | Duplicate_Count |
|----------------|-----------------|
| CA-2016-152156 | 2               |
| US-2015-108966 | 2               |
| CA-2014-115812 | 7               |
| US-2015-118983 | 2               |
| CA-2014-143336 | 3               |
| CA-2016-137330 | 2               |
| CA-2016-121755 | 2               |
| US-2015-150630 | 6               |
| CA-2016-117990 | 2               |

Result 14 x

Output

Action Output

| #  | Time     | Action                                                                  | Message              | Duration / Fetch      |
|----|----------|-------------------------------------------------------------------------|----------------------|-----------------------|
| 34 | 00:06:03 | SELECT 'Customer Name', SUM(Sales) AS Total_Sales FROM Superstore GR... | 10 row(s) returned   | 0.031 sec / 0.000 sec |
| 35 | 00:06:51 | SELECT 'Order ID', COUNT(*) AS Duplicate_Count FROM Superstore GROUP... | 1000 row(s) returned | 0.032 sec / 0.000 sec |

**Solve use cases (monthly trends, top customers, duplicates)**

MySQL Workbench

New Connection x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- City
- State
- Postal Code
- Region
- Product ID
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit
- Indexes

Administration Schemas

Information

Table: superstore

Columns:

- Row ID int
- Order ID text
- Order Date text
- Ship Date text
- Ship Mode text
- Customer text
- ID text
- Customer text
- Name text

Object Info Session

SQL File 3

Limit to 1000 rows

```

69 HAVING COUNT(*) > 1;
70
71 SELECT YEAR('Order Date') AS Year,
72        MONTH('Order Date') AS Month,
73        SUM(Sales) AS Monthly_Sales
74 FROM Superstore
75 GROUP BY YEAR('Order Date'),

```

Result Grid

| Year | Month | Monthly_Sales      |
|------|-------|--------------------|
| 2016 | 1     | 2272449.8562999545 |

Result 15 x

Output

Action Output

| #  | Time     | Action                                                                                                                                                         | Message              | Duration / Fetch      |
|----|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------|
| 35 | 00:06:51 | SELECT 'Order ID', COUNT(*) AS Duplicate_Count FROM Superstore GROUP BY Order ID;                                                                              | 1000 row(s) returned | 0.032 sec / 0.000 sec |
| 36 | 00:07:53 | SELECT YEAR('Order Date') AS Year, MONTH('Order Date') AS Month, SUM(Sales) AS Monthly_Sales FROM Superstore GROUP BY YEAR('Order Date'), MONTH('Order Date'); | 1 row(s) returned    | 0.031 sec / 0.000 sec |

MySQL Workbench

New Connection x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- City
- State
- Postal Code
- Region
- Product ID
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit
- Indexes

Administration Schemas

Information

Table: superstore

Columns:

- Row ID int
- Order ID text
- Order Date text
- Ship Date text
- Ship Mode text
- Customer text
- ID text
- Customer text
- Name text

Object Info Session

SQL File 3

Limit to 1000 rows

```

77 ORDER BY Year, Month;
78
79 SELECT Region,
80        SUM(Sales) AS Total_Sales
81 FROM Superstore
82 GROUP BY Region;

```

Result Grid

| Region  | Total_Sales        |
|---------|--------------------|
| South   | 388983.58500000037 |
| West    | 713471.34450000004 |
| Central | 497800.87280000007 |
| East    | 672194.05399999981 |

Result 16 x

Output

Action Output

| #  | Time     | Action                                                                                                                                                         | Message           | Duration / Fetch      |
|----|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------------------|
| 36 | 00:07:53 | SELECT YEAR('Order Date') AS Year, MONTH('Order Date') AS Month, SUM(Sales) AS Monthly_Sales FROM Superstore GROUP BY YEAR('Order Date'), MONTH('Order Date'); | 1 row(s) returned | 0.031 sec / 0.000 sec |
| 37 | 00:08:36 | SELECT Region, SUM(Sales) AS Total_Sales FROM Superstore GROUP BY Region;                                                                                      | 4 row(s) returned | 0.015 sec / 0.000 sec |

MySQL Workbench

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- City
- State
- Postal Code
- Region
- Product ID
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit
- Indexes

Administration Schemas

Information

Table: superstore

Columns:

- Row ID int
- Order ID text
- Order Date text
- Ship Date text
- Ship Mode text
- Customer text
- ID text
- Customer text
- Name text

Object Info Session

SQL File 3\*

Limit to 1000 rows

```
82 GROUP BY Region;
83
84 SELECT Category,
85 SUM(Profit) AS Total_Profit
86 FROM Superstore
87 GROUP BY Category;
```

Result Grid

| Category        | Total_Profit       |
|-----------------|--------------------|
| Furniture       | 16980.77119999998  |
| Office Supplies | 120489.88640000001 |
| Technology      | 145387.09659999999 |

Result 17

Output

Action Output

| #  | Time     | Action                                                                    | Message           | Duration / Fetch      |
|----|----------|---------------------------------------------------------------------------|-------------------|-----------------------|
| 37 | 00:08:36 | SELECT Region, SUM(Sales) AS Total_Sales FROM Superstore GROUP BY ...     | 4 row(s) returned | 0.015 sec / 0.000 sec |
| 38 | 00:09:41 | SELECT Category, SUM(Profit) AS Total_Profit FROM Superstore GROUP BY ... | 3 row(s) returned | 0.031 sec / 0.000 sec |

Validate results (row counts, data quality).

MySQL Workbench

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- City
- State
- Postal Code
- Region
- Product ID
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit
- Indexes

Administration Schemas

Information

Table: superstore

Columns:

- Row ID int
- Order ID text
- Order Date text
- Ship Date text
- Ship Mode text
- Customer text
- ID text
- Customer text
- Name text

Object Info Session

SQL File 3\*

Limit to 1000 rows

```
87 GROUP BY Category;
88
89 SELECT Region,
90 COUNT(*) AS Orders_Count,
91 SUM(Sales) AS Revenue,
92 SUM(Profit) AS Profit
93 FROM Superstore
```

Result Grid

| Region  | Orders_Count | Revenue            | Profit             |
|---------|--------------|--------------------|--------------------|
| West    | 3099         | 713471.34450000004 | 106021.14950000007 |
| East    | 2756         | 672194.05399999981 | 90672.01270000022  |
| Central | 2260         | 497800.87280000007 | 40128.903          |
| South   | 1579         | 388983.58500000037 | 46035.689000000004 |

Result 18

Output

Action Output

| #  | Time     | Action                                                                    | Message           | Duration / Fetch      |
|----|----------|---------------------------------------------------------------------------|-------------------|-----------------------|
| 38 | 00:09:41 | SELECT Category, SUM(Profit) AS Total_Profit FROM Superstore GROUP BY ... | 3 row(s) returned | 0.031 sec / 0.000 sec |
| 39 | 00:10:53 | SELECT Region, COUNT(*) AS Orders_Count, SUM(Sales) AS Revenue, ...       | 4 row(s) returned | 0.031 sec / 0.000 sec |



MySQL Workbench

New Connection x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- City
- State
- Postal Code
- Region
- Product ID
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit
- Indexes

Administration Schemas

Information

Table: superstore

Columns:

- Row ID int
- Order ID text
- Order Date text
- Ship Date text
- Ship Mode text
- Customer ID text
- ID text
- Customer Name text

Object Info Session

SQL File 3'

Limit to 1000 rows

```

95 ORDER BY Revenue DESC;
96
97 SELECT *
98 FROM Superstore
99 WHERE Sales IS NULL
100 OR Quantity IS NULL
101 OR Region IS NULL;

```

Result Grid

| Row ID | Order ID | Order Date | Ship Date | Ship Mode | Customer ID | Customer Name | Segment | Country | City | State | Postal Code | Region | Product ID | Category | Sub-Category |
|--------|----------|------------|-----------|-----------|-------------|---------------|---------|---------|------|-------|-------------|--------|------------|----------|--------------|
|--------|----------|------------|-----------|-----------|-------------|---------------|---------|---------|------|-------|-------------|--------|------------|----------|--------------|

Superstore 19 x

Output

Action Output

| #  | Time     | Action                                                                | Message           | Duration / Fetch      |
|----|----------|-----------------------------------------------------------------------|-------------------|-----------------------|
| 39 | 00:10:53 | SELECT Region, COUNT(*) AS Orders_Count, SUM(Sales) AS Revenue, ...   | 4 row(s) returned | 0.031 sec / 0.000 sec |
| 40 | 00:11:29 | SELECT * FROM Superstore WHERE Sales IS NULL OR Quantity IS NULL O... | 0 row(s) returned | 0.015 sec / 0.000 sec |

MySQL Workbench

New Connection x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- City
- State
- Postal Code
- Region
- Product ID
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit
- Indexes

Administration Schemas

Information

Table: superstore

Columns:

- Row ID int
- Order ID text
- Order Date text
- Ship Date text
- Ship Mode text
- Customer ID text
- ID text
- Customer Name text

Object Info Session

SQL File 3'

Limit to 1000 rows

```

99 WHERE Sales IS NULL
100 OR Quantity IS NULL
101 OR Region IS NULL;
102
103 SELECT COUNT(*) AS Total_Rows
104 FROM Superstore;

```

Result Grid

| Row ID | Order ID | Order Date | Ship Date | Ship Mode | Customer ID | Customer Name | Segment | Country | City | State | Postal Code | Region | Product ID | Category | Sub-Category |
|--------|----------|------------|-----------|-----------|-------------|---------------|---------|---------|------|-------|-------------|--------|------------|----------|--------------|
|--------|----------|------------|-----------|-----------|-------------|---------------|---------|---------|------|-------|-------------|--------|------------|----------|--------------|

Total\_Rows

9694

Result 20 x

Output

Action Output

| #  | Time     | Action                                                                | Message           | Duration / Fetch      |
|----|----------|-----------------------------------------------------------------------|-------------------|-----------------------|
| 40 | 00:11:29 | SELECT * FROM Superstore WHERE Sales IS NULL OR Quantity IS NULL O... | 0 row(s) returned | 0.015 sec / 0.000 sec |
| 41 | 00:12:20 | SELECT COUNT(*) AS Total_Rows FROM Superstore LIMIT 0, 1000           | 1 row(s) returned | 0.000 sec / 0.000 sec |